

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the [full course website](#) for more.

- [Intro to Pandas DataFrame](#)
- [Intro to RAPIDS cuDF to accelerate pandas](#)
- [Getting Started with cuML's accelerator mode](#)

Using Accelerated Hardware

- [Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API](#)
- [Train a Vision Transformer \(ViT\) for image classification with JAX](#)
- [Text classification with a transformer language model using JAX](#)

✓ Featured examples

- [Train a miniGPT language model with JAX AI Stack](#)
- [LoRA/QLoRA finetuning for LLM using Tunix](#)
- [Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA](#)
- [Loading Hugging Face Transformers Checkpoints](#)
- [8-bit Integer Quantization in Keras](#)
- [Float8 training and inference with a simple Transformer model](#)
- [Pretraining a Transformer from scratch with KerasHub](#)
- [Simple MNIST convnet](#)
- [Image classification from scratch using Keras 3](#)
- [Image Classification with KerasHub](#)

```
!apt-get update -qq
!apt-get install -y -qq r-base
```

```
W: Skipping acquire of configured file 'main/source/Sources' as repository 'http
(Reading database ... 118243 files and directories currently installed.)
Preparing to unpack .../r-base-core_4.6.1-1.2204.1_amd64.deb ...
Unpacking r-base-core (4.6.1-1.2204.1) over (4.6.0-4.2204.0) ...
Preparing to unpack .../r-base_4.6.1-1.2204.1_all.deb ...
Unpacking r-base (4.6.1-1.2204.1) over (4.6.0-4.2204.0) ...
Preparing to unpack .../r-recommended_4.6.1-1.2204.1_all.deb ...
Unpacking r-recommended (4.6.1-1.2204.1) over (4.6.0-4.2204.0) ...
Setting up r-base-core (4.6.1-1.2204.1) ...
Installing new version of config file /etc/R/Makeconf ...
Setting up r-recommended (4.6.1-1.2204.1) ...
Setting up r-base (4.6.1-1.2204.1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
```

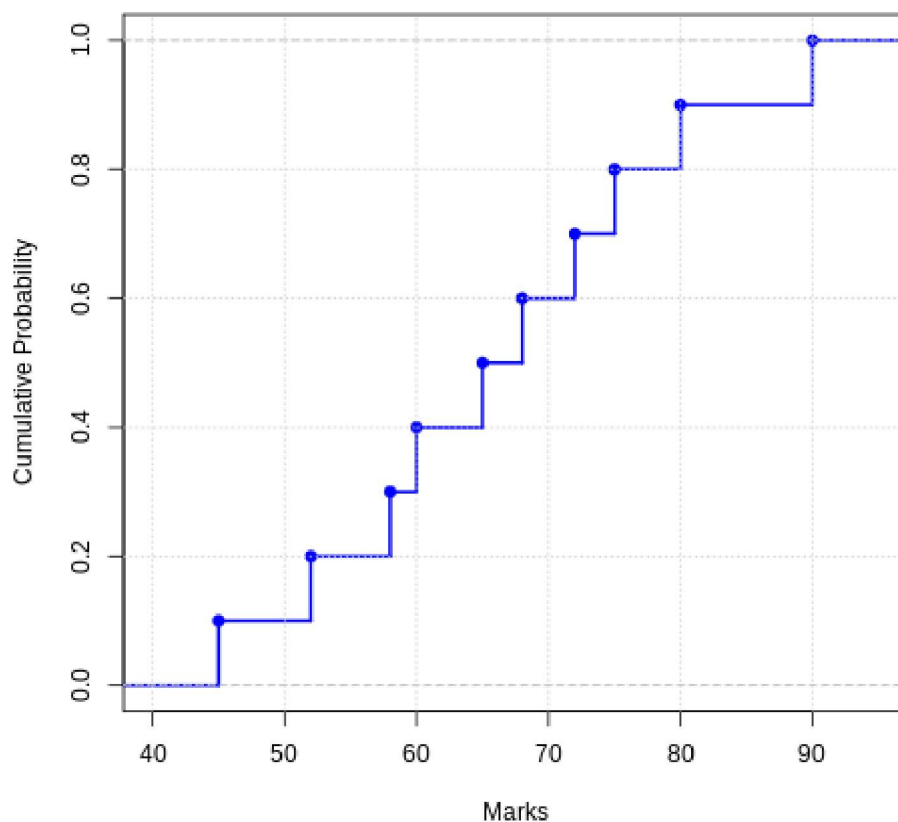
```
Processing triggers for man-db (2.10.2-1) ...  
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
```

```
!pip install rpy2  
%load_ext rpy2.ipython
```

```
Requirement already satisfied: rpy2 in /usr/local/lib/python3.12/dist-packages (  
Requirement already satisfied: cffi>=1.15.1 in /usr/local/lib/python3.12/dist-pa  
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages  
Requirement already satisfied: tzlocal in /usr/local/lib/python3.12/dist-package  
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packa  
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist  
The rpy2.ipython extension is already loaded. To reload it, use:  
%reload_ext rpy2.ipython
```

```
%%R  
  
marks <- c(45,52,58,60,65,68,72,75,80,90)  
  
ecdf_marks <- ecdf(marks)  
  
result <- data.frame(  
  Marks=sort(marks),  
  Cumulative_Probability=ecdf_marks(sort(marks))  
)  
  
print(result)  
  
plot(ecdf_marks,  
  main="ECDF of Student Marks",  
  xlab="Marks",  
  ylab="Cumulative Probability",  
  verticals=TRUE,  
  do.points=TRUE,  
  col="blue",  
  lwd=2)  
  
grid()
```

	Marks	Cumulative_Probability
1	45	0.1
2	52	0.2
3	58	0.3
4	60	0.4
5	65	0.5
6	68	0.6
7	72	0.7
8	75	0.8
9	80	0.9
10	90	1.0

ECDF of Student Marks

```

; <- c(4200,5000,5600,6100,6800,7200,8100,9200,10400,12000)

_steps <- ecdf(steps)

ecdf_steps,
:"ECDF of Daily Steps",
:"Daily Steps",
:"Cumulative Probability",
.cals=TRUE,
)ints=TRUE,
'darkgreen",

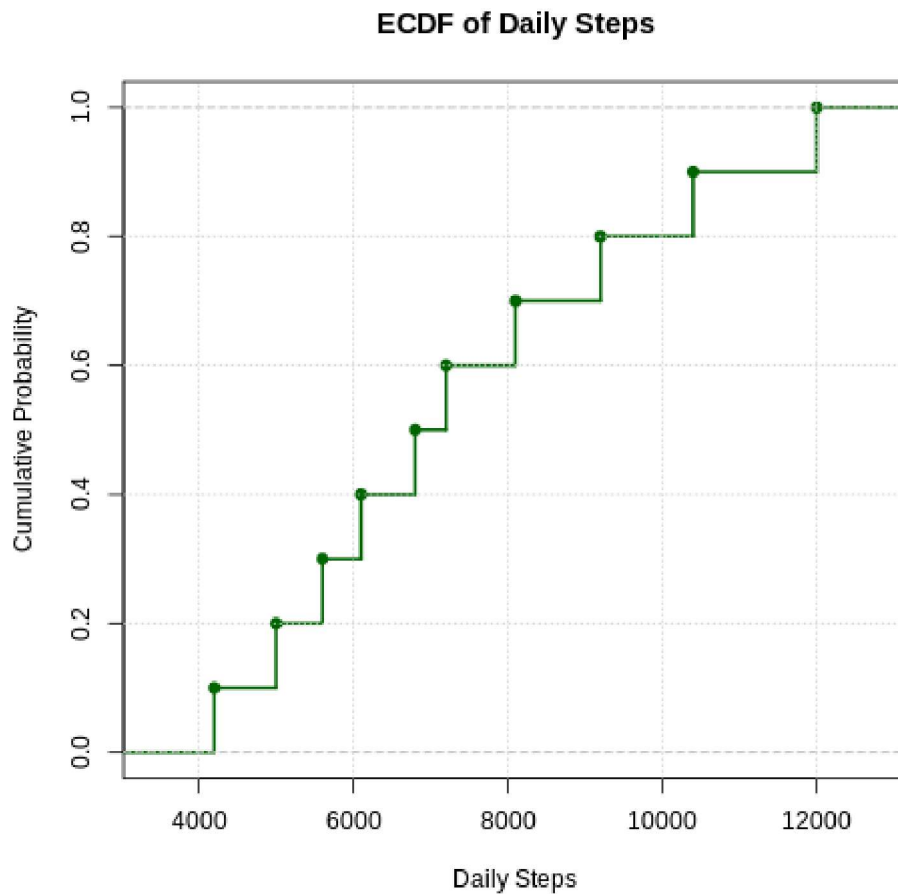
```

```
!)\n\n)

proportion <- ecdf_steps(8000)

'Proportion of participants with 8000 steps or fewer =', proportion, "\n")
```

Proportion of participants with 8000 steps or fewer = 0.6



```
%%R

purchase <- c(250,300,320,350,380,450,520,650,1800,5200)

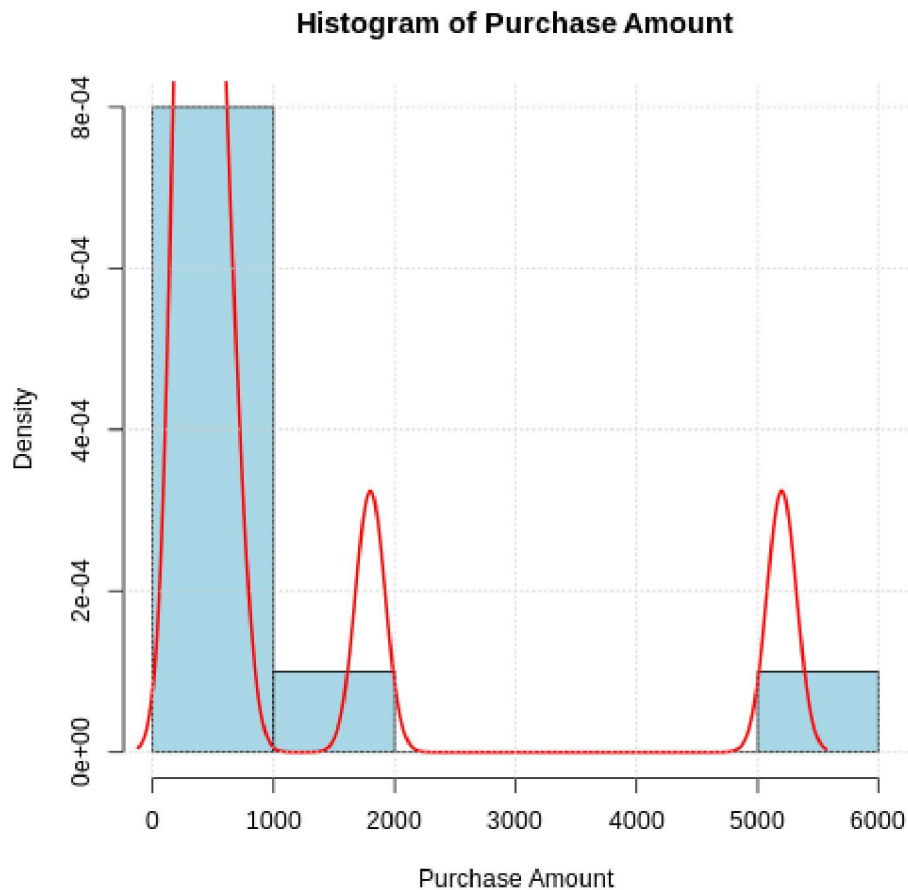
hist(purchase,
probability=TRUE,
main="Histogram of Purchase Amount",
xlab="Purchase Amount",
col="lightblue",
border="black")

lines(density(purchase),
col="red",
lwd=2)
```

```
grid()

cat("Distribution is Positively Skewed\n")
```

Distribution is Positively Skewed



```
%%R

waiting <- c(12,15,18,20,22,24,25,28,30,32)

qqnorm(waiting,
main="Normal Q-Q Plot of Waiting Time",
pch=19,
col="blue")

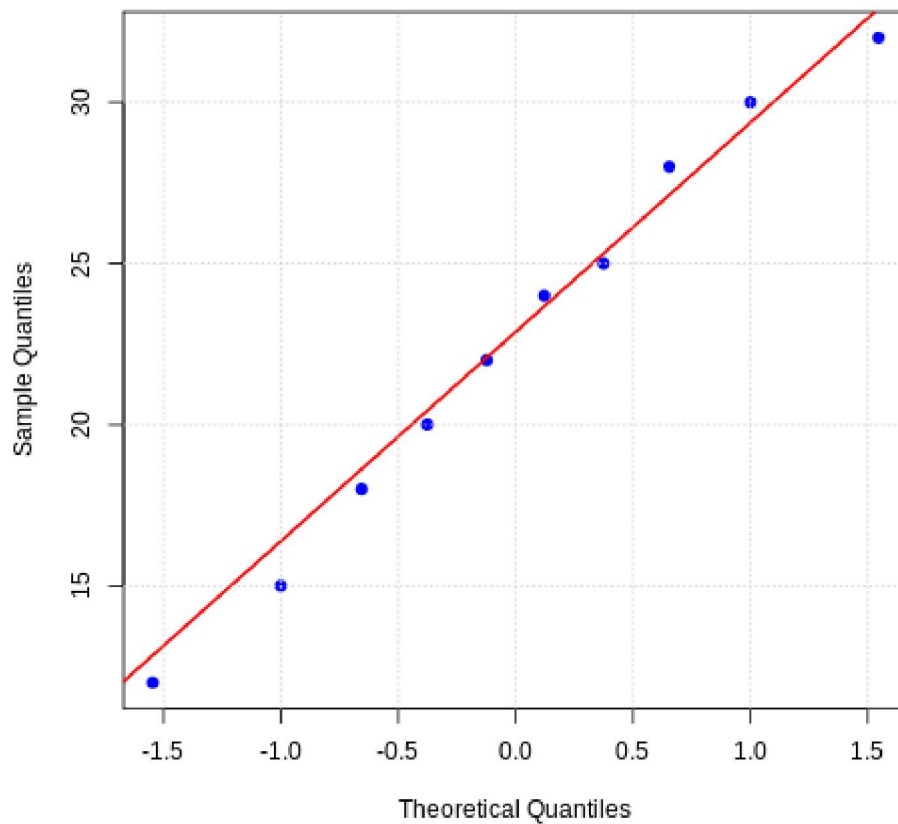
qqline(waiting,
col="red",
lwd=2)

grid()

cat("Waiting times approximately follow a normal distribution.\n")
```

Waiting times approximately follow a normal distribution.

Normal Q-Q Plot of Waiting Time



```
income <- c(18000, 22000, 24000, 26000, 28000,
            34000, 42000, 85000, 200000)
```

```
plot(income,
      main="Histogram of Monthly Income",
      xlab="Monthly Income",
      col="lightgreen",
      border="black")
```

```
plot(density(income),
      main="Density Plot of Monthly Income",
      col="blue",
```

```
plot(income,
      main="Normal Q-Q Plot of Monthly Income",
      col="darkgreen")
```

Income data is highly positively skewed and not normally distributed.

